

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

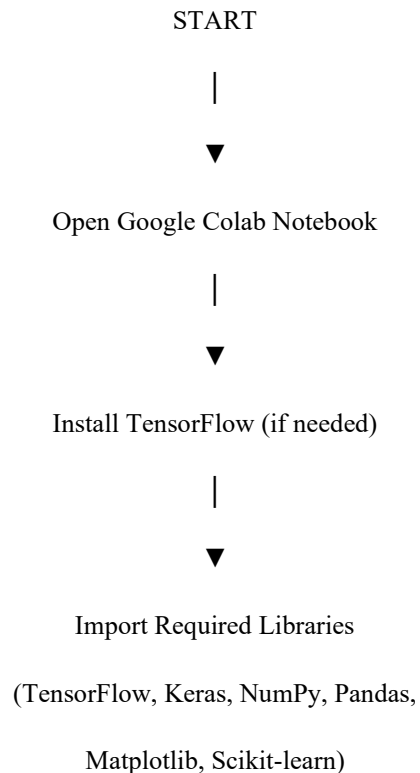
Division: E Batch: 3 Practical No: 1

Roll no: 05 PRN No: 12411040 Name of Student: **Prathmesh Patil**

Subject Name : Deep Learning

Problem Statement : Install and configure TensorFlow/Keras in Google Colab.
Perform data preprocessing, normalization, train-test split, and visualization on a sample dataset.

Algorithm :



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Load Sample Dataset (Iris)

|



Perform Data Preprocessing
(Separate Features and Target)

|



Normalize the Dataset
(Using StandardScaler)

|



Split Data into Train & Test
(80% Training, 20% Testing)

|



Visualize the Dataset
(Scatter Plot using Matplotlib)

|



Verify TensorFlow Installation
(Display Version Number)

|



END

Code :

```
import tensorflow as tf

mnist = tf.keras.datasets.mnist

(X_train, y_train), (X_test, y_test) = mnist.load_data()

print("Training Images:", X_train.shape)
print("Training Labels:", y_train.shape)
print("Testing Images:", X_test.shape)
print("Testing Labels:", y_test.shape)

import matplotlib.pyplot as plt

plt.imshow(X_train[0], cmap='gray')
plt.title(f"Label: {y_train[0]}")
plt.axis('off')
plt.show()

X_train = X_train / 255.0
X_test = X_test / 255.0

print("Minimum Pixel Value:", X_train.min())
print("Maximum Pixel Value:", X_train.max())
```

```
plt.figure(figsize=(10,5))

for i in range(10):

    plt.subplot(2,5,i+1)

    plt.imshow(X_train[i], cmap='gray')

    plt.title(f"Label: {y_train[i]} ")

    plt.axis('off')

plt.show()


print(X_train.dtype)

print(X_train.min())

print(X_train.max())

print("Minimum:", X_train.min())

print("Maximum:", X_train.max())


print(X_train.shape)

print(X_test.shape)


plt.hist(X_train.flatten(), bins=50)

plt.title("Distribution of Pixel Values")

plt.xlabel("Pixel Value")

plt.ylabel("Frequency")

plt.show()

import numpy as np

unique, counts = np.unique(y_train, return_counts=True)
```

```
print(dict(zip(unique, counts)))
```

```
plt.bar(unique, counts)
```

```
plt.xlabel("Digit")
```

```
plt.ylabel("Number of Images")
```

```
plt.title("Distribution of Digits in MNIST")
```

```
plt.show()
```

Output :

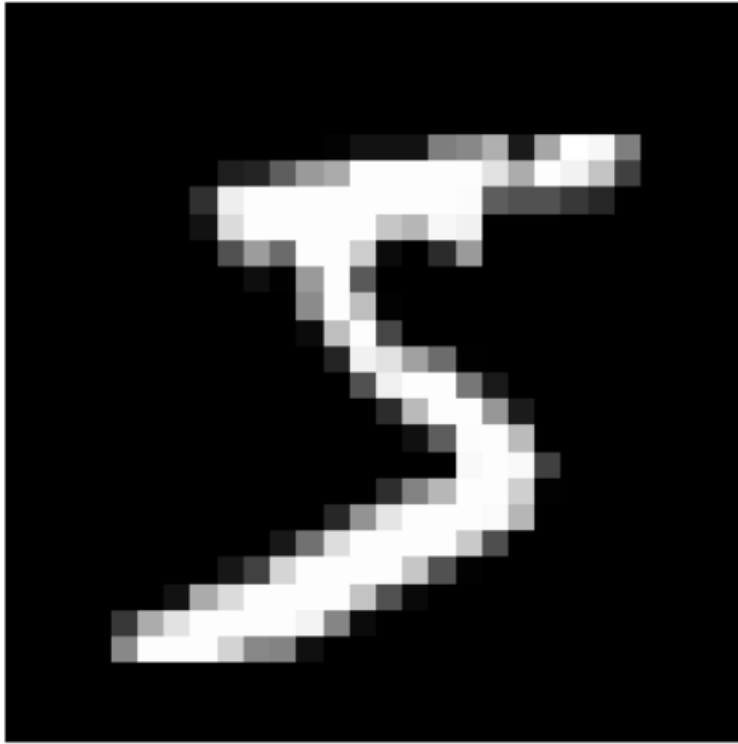
Training Images: (60000, 28, 28)

Training Labels: (60000,)

Testing Images: (10000, 28, 28)

Testing Labels: (10000,)

Label: 5



Minimum Pixel Value: 0.0

Maximum Pixel Value: 1.0

